

PROJECT DOCUMENTATION

AI-Powered Customer Support & Service Automation

*A Salesforce Service Cloud, Flow Automation & Agentforce-Based Customer Support Solution for an
International Electronics & Gadgets Company*

Prepared using Salesforce Developer Edition • Flow Builder • Agentforce

1. Project Overview

The AI-Powered Customer Support & Service Automation project is a Salesforce-based application built for an international Electronics & Gadgets company to streamline its customer service operations. The system automates the handling of product issues, order tracking requests, warranty claims, and general support inquiries using Salesforce Service Cloud, record-triggered Flows, and Agentforce (AI Service Agent).

The solution enables support teams to efficiently manage customers, products, orders, cases, queues, and approvals — giving customers faster, more consistent responses and giving management real-time visibility into case resolution performance, all within a single unified Salesforce environment.

2. Problem Statement

The company currently handles a high volume of customer inquiries manually. This leads to delayed responses, inconsistent service quality, and limited visibility into how efficiently cases are resolved. A centralized, automated system was required to classify incoming requests, assign priority and SLA deadlines, route cases to the correct queue, support agents with AI-driven guidance, and give management clear performance reporting.

3. Skills Required

- Salesforce (Service Cloud)
- Agentforce
- Prompt Builder
- Data Modelling
- Flow Designer & Process Automation
- Data Validation

4. Requirement Analysis & Planning

This phase focused on understanding business needs and defining the project scope before development began. Key objectives were identified, including automated case classification and routing, SLA-based priority handling, and AI-based agent assistance using Salesforce Agentforce and Prompt Builder.

Business requirements were analyzed to determine the necessary Salesforce objects, fields, relationships, and security settings. Based on these requirements, a structured project plan was prepared, outlining milestones, timelines, and deliverables — ensuring the project was well-planned, technically feasible, and aligned with business goals before development started.

- Centralized customer case management for product, order and warranty issues.
- Automatic case classification (Damage / Warranty) based on issue details.
- Priority assignment and SLA-based case reassignment.
- Agent assistance through Agentforce AI suggestions.
- Role-based access control and field-level security.
- Reports and dashboards for support performance monitoring.

5. System Architecture

The architecture represents the complete support lifecycle, from the moment a customer raises a request to case resolution and continuous improvement.

Customer Inquiry → Case Creation → Classification (Damage / Warranty) → Priority & SLA Assignment → Queue Routing → Agentforce AI Assistance → Agent Resolution → Reports & Feedback

Incoming requests are captured as User Case records. Record-triggered Flows classify each case and route it to the appropriate queue, while a second automation evaluates priority and reassigns high-priority cases to keep SLA commitments on track. Agentforce supports agents with AI-generated guidance during resolution, and reports feed performance data back to management.

6. Functional Scope

The functional scope of this project covers an automated, AI-assisted customer support and warranty-claims system built on Salesforce. The application allows the creation and management of customer case records, capturing details such as case type (Damage or Warranty), priority, status, and owner.

Record-triggered Flows automatically classify each case, assign it to the correct queue (Damage Queue or Warranty Queue), and update case ownership. A second flow evaluates case priority and automatically reassigns high-priority cases for faster handling, helping the business meet SLA commitments.

The system provides AI-powered assistance through Salesforce Agentforce, giving support agents suggested responses and guidance during case resolution. A dedicated Lightning App brings these capabilities together in a single, purpose-built interface for support staff, and the solution supports secure data handling through role-based access, field-level security, and validation rules to maintain data accuracy and integrity.

7. Stakeholder Mapping

- **Project Sponsor:** Defines business objectives and approves deliverables.
- **Salesforce System Administrator:** Configures objects, Flows, security settings, and Agentforce; manages system maintenance.
- **Support Agents:** Primary end users — manage assigned cases, apply AI-generated suggestions, and resolve customer issues.
- **Support Managers:** Monitor SLA compliance, queue distribution, and agent performance.
- **Business / Management Stakeholders:** Define requirements, validate business objectives, and use reports for decision-making.
- **IT Support:** Ensures ongoing system maintenance and resolves technical issues.

8. Execution Roadmap

- **Phase 1 — Requirement Analysis & Planning:** Business requirements were gathered, AI use cases were identified, and the overall project scope and timeline were defined.
- **Phase 2 — Backend Development & Configuration:** Custom objects, fields, relationships, and validation rules were created for case management. Record-triggered Flows were configured for case classification, queue routing, and SLA/priority handling.

- **Phase 3 — UI/UX Development & Customization:** Lightning Record Pages were customized and a dedicated Lightning App was built using Lightning App Builder, integrating AI-enabled fields for agents.
- **Phase 4 — Data Migration, Testing & Security:** Sample and test data were migrated; functional testing validated Flow automation, case-classification accuracy, and system behaviour. Profile-based access and field-level security were applied.
- **Phase 5 — Deployment, Documentation & Maintenance:** Final configurations were activated, project documentation was prepared, and maintenance / future-enhancement plans were defined.

9. System Requirements

Software

- Salesforce Developer Org
- Lightning Experience
- Salesforce Flow Builder
- Agentforce / Prompt Builder
- Reports & Dashboards

Functional Requirements

- Case management
- SLA & priority tracking
- Classification automation
- AI-based agent assistance
- Security controls

Non-Functional Requirements

- Scalability
- Data accuracy
- Reliability
- Easy maintenance

10. Salesforce Environment Setup

A Salesforce Developer Edition org was created to build and test the solution in a free, full-featured environment.

Activity 1: Creating a Developer Account

- Go to developer.salesforce.com/signup.
- Enter First Name, Last Name, Email, Role (Developer), Company (organization name), Country, and Postal Code.
- Choose a Username in the format `username@organization.com` (this does not need to be a real email address).
- Click Sign Me Up.

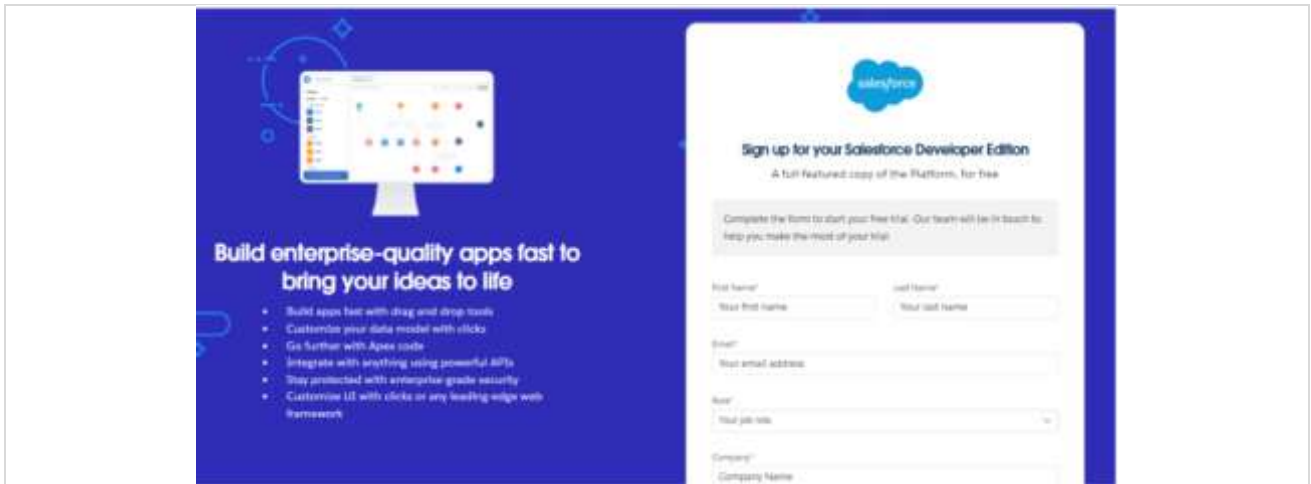


Figure 1: Salesforce Developer Edition sign-up form.

Activity 2: Account Activation

- Open the inbox of the email address used to sign up (delivery may take 5–10 minutes).
- Open the “Welcome to Salesforce: Verify your account” email and click Verify Account.

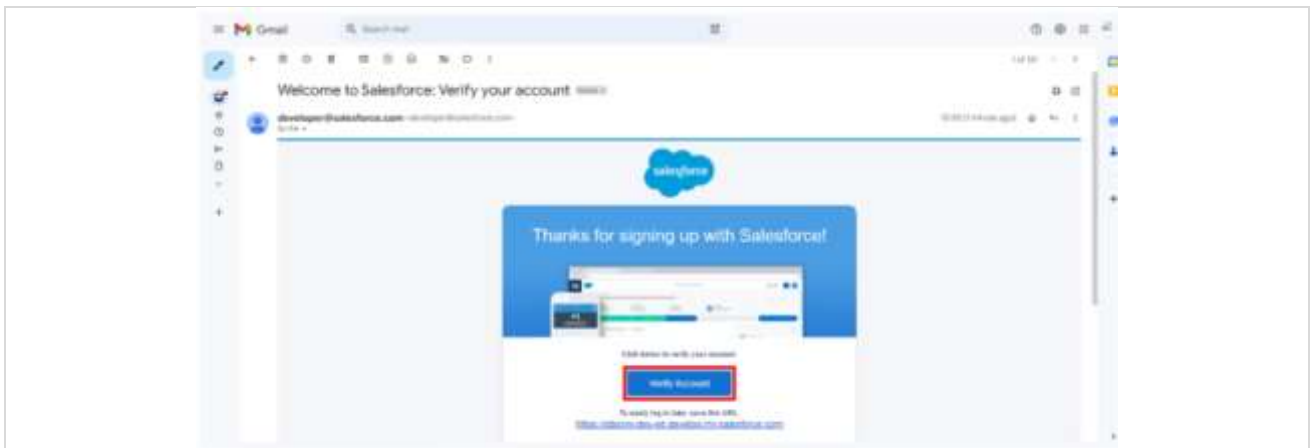


Figure 2: Account-verification email.

- Set a new password that meets the minimum requirements (8 characters, 1 letter, 1 number), choose a security question, and click Change Password.

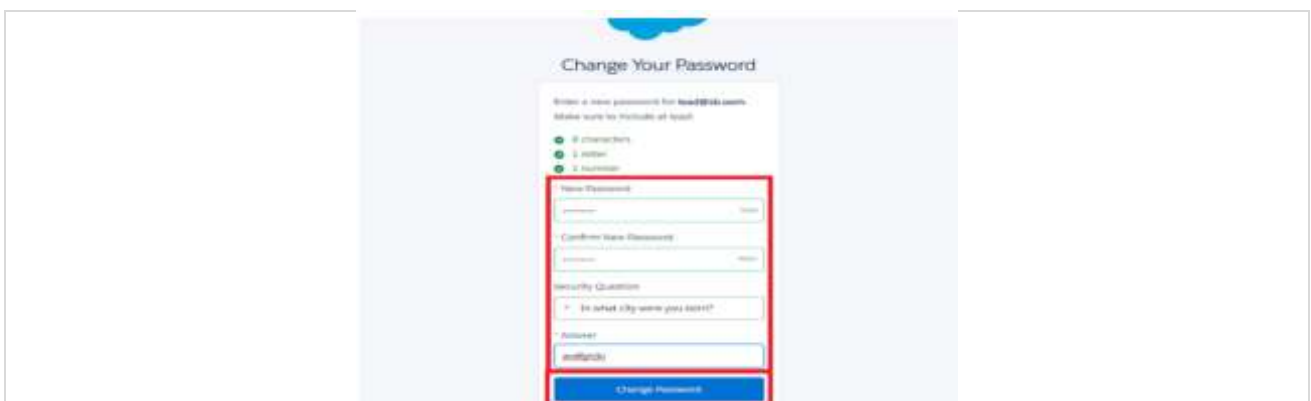


Figure 3: Setting a new password on first login.

Once logged in, the Salesforce Setup Home page becomes the starting point for all configuration work carried out in this project.

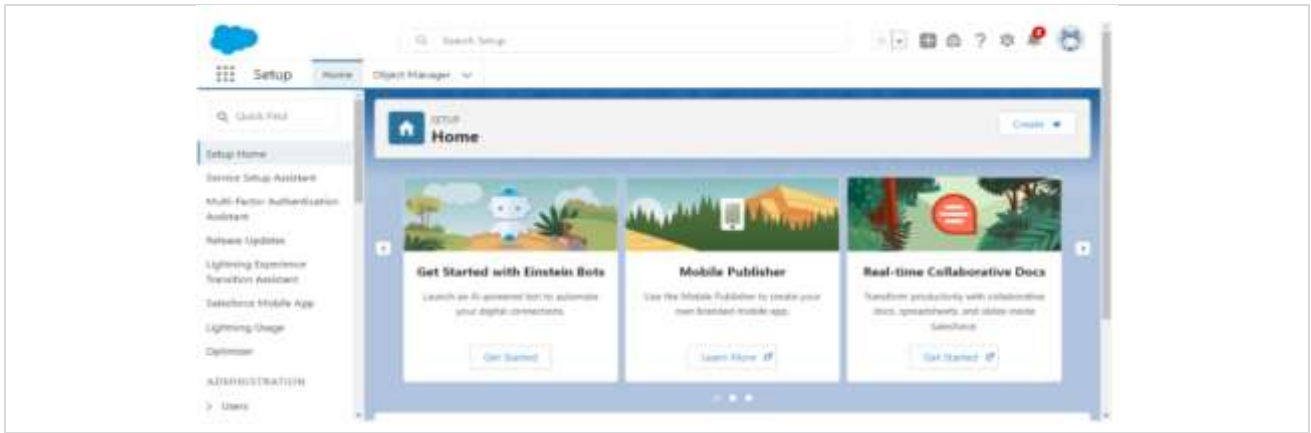


Figure 4: Salesforce Setup Home page.

11. Data Architecture & Object Model

Standard Objects

- **Account** — Stores customer / organization information.
- **Contact** — Stores individual customer contact details.

Custom Object

- **User_Case__c ("User Case")** — Stores each customer support / warranty case raised against a product or order.
- **Customers__c("Customer")** — Stores Customer details.
- **Mechendices__c(Merchendice")** — Gives the products details and wishlist of product.

Key Fields

- **Case_Type__c** — Damage or Warranty — drives automatic queue routing.
- **Priority__c** — Low / Medium / High / Critical — drives SLA reassignment.
- **CaseStatus__c** — Open / Closed — tracks the current state of the case.
- **OwnerId** — Assigned queue or agent responsible for the case.

Relationships

- Account → User_Case__c (a case is raised by a customer account).
- User_Case__c → Queue (Damage Queue / Warranty Queue) — based on Case Type.

12. Automation Logic — Flow Implementation

Two record-triggered Flows automate the core support process: one classifies and routes new or updated cases, and a second re-assigns high-priority cases to help the team meet SLA targets.

12.1 Case Classification & Routing Flow

- In Setup, search for Flows, click New Flow, select Record-Triggered Flow, and click Create.
- Object: User Case. Trigger the Flow When: A record is created or updated. Entry Conditions: None. Optimize for: Actions and Related Records.
- Add an Assignment element labelled Case S1 that sets CaseStatus to Open.

- Add a Decision element labelled Check Case Type with two outcomes — Damage Case (Case Type equals Damage) and Warranty Case (Case Type equals Warranty).
- For each outcome, use Get Records to retrieve the matching queue (Get Damage Queue / Get Warranty Queue), then Update Records to assign the case to that queue.

Configure Start

Select Object:
Select the object whose records trigger the flow when they're created, updated, or deleted.
* Object:

Configure Trigger:
Trigger the Flow When:
☐ A record is created
☐ A record is updated
☒ A record is created or updated
☐ A record is deleted

Set Entry Conditions:
Specify entry conditions to reduce the number of records that trigger the flow and the number of times the flow is executed. Minimizing unnecessary flow executions helps to conserve your org's resources.
If you create a flow that's triggered when a record is updated, we recommend first defining entry conditions. Then select the **Only when a record is updated to meet the condition requirements** option for When to Run the Flow for Updated Records.
Condition Requirements:

Figure 5: Configuring the record-triggered flow start on the User Case object.

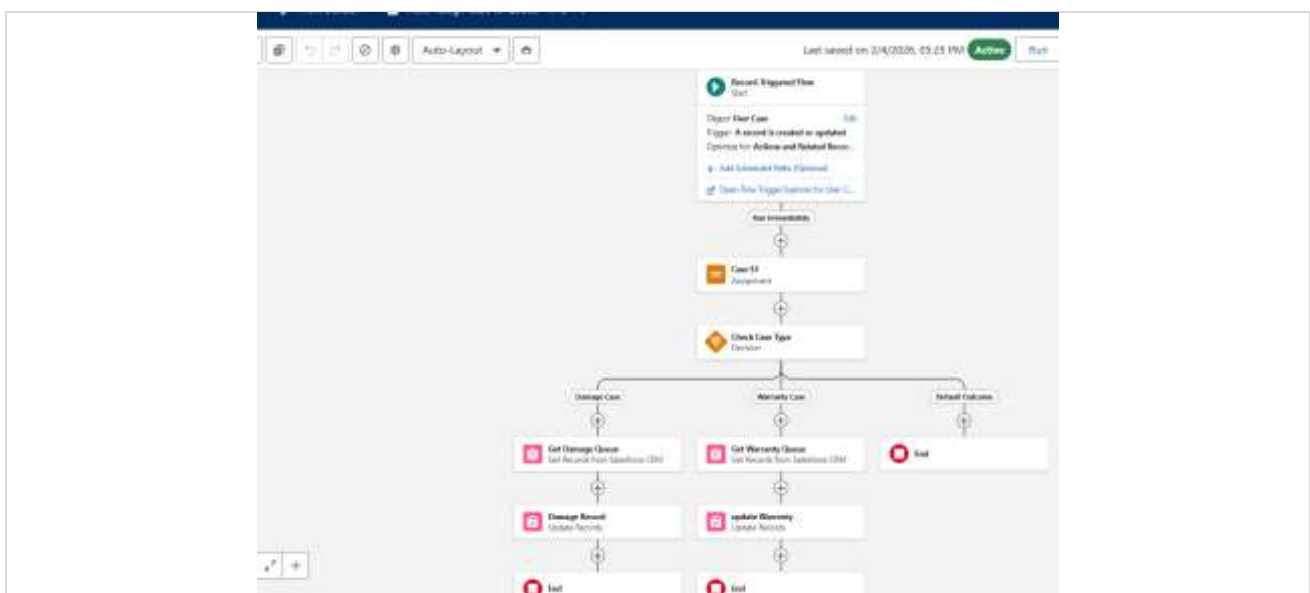


Figure 6: Completed Case Classification & Routing Flow — Active.

12.2 Case Auto Assignment & SLA Flow

- A second Record-Triggered Flow is created on the User Case object, this time triggering only When: A record is created.
- Add a Decision element labelled Is High Priority, evaluating `{!$Record.Priority__c} Equals High`.
- Under this outcome, add Update Records to reassign the case OwnerId, ensuring high-priority cases reach an agent immediately.
- The flow is saved as Case Auto Assignment & SLA Flow and Activated.

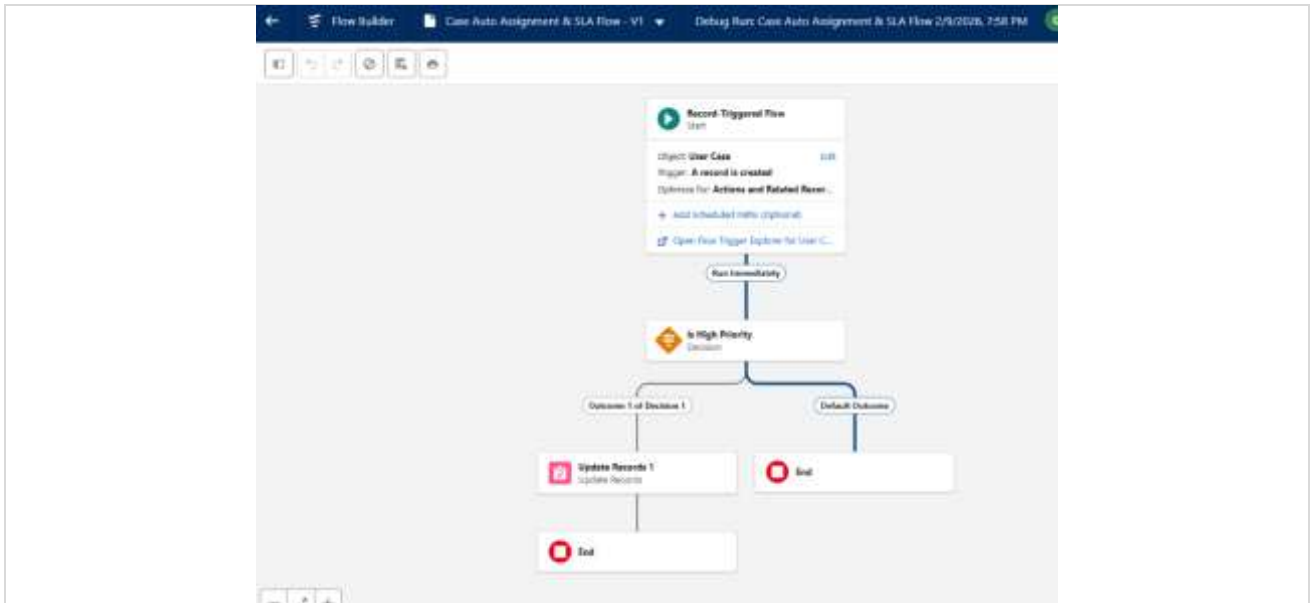


Figure 7: Case Auto Assignment & SLA Flow — Is High Priority decision (Debug Run).

13. Lightning App Configuration

A dedicated Lightning App brings the case-management objects and support tools together in a single, purpose-built workspace for agents.

- In Setup, search for App Manager and click New Lightning App.
- App Name: AI-Powered Customer Support & Service Automation. Developer Name: auto-populated. Description: a short, meaningful summary of the app. Image and primary colour: left as default (optional).
- Click Next through App Options and Utility Items, keeping the defaults.
- On Navigation Items, search for and add the relevant items — including the User Case custom object created earlier — using the arrow button, then click Next to finish.

14. Agentforce AI Integration

With core case management and SLA automation in place, this phase layers AI-powered assistance on top using Salesforce Agentforce and Prompt Builder. The AI Service Agent is designed to support agents with suggested responses and similar-case guidance generated from case details such as type, priority, and description — reducing manual effort in the same way the classification Flows already do for routing.

- Suggested responses / actions for agents while working a case.
- Similar-case guidance drawn from case history.
- Feedback capture from agents on suggestion quality.
- Continuous improvement of AI responses through feedback analysis.

15. Reports & Monitoring

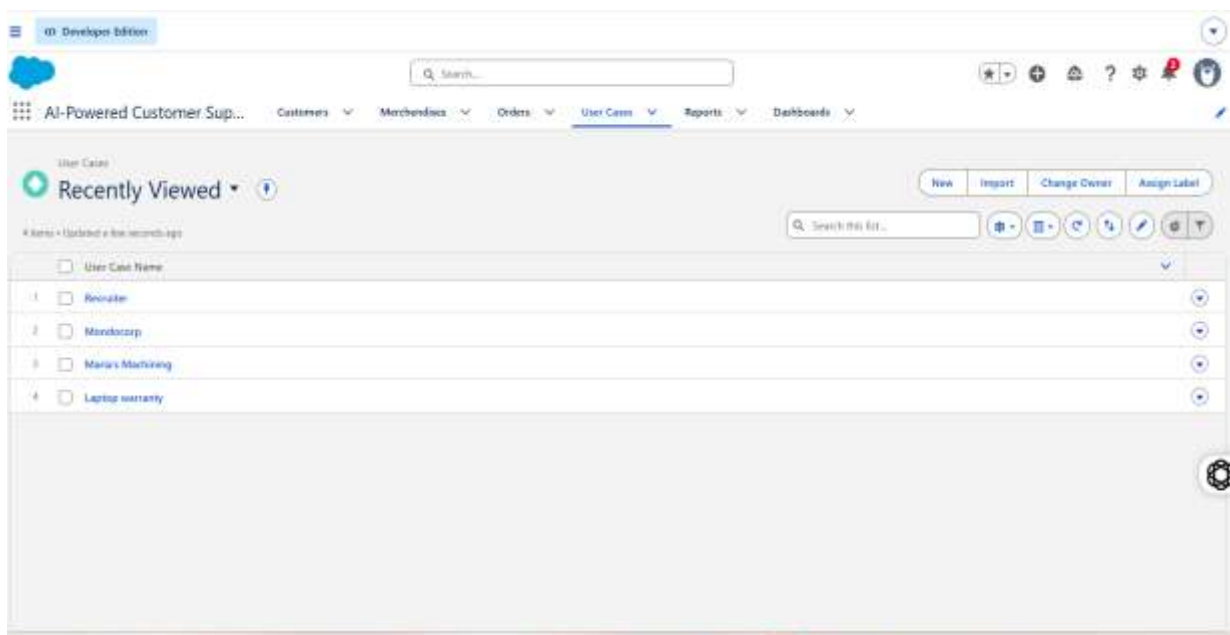
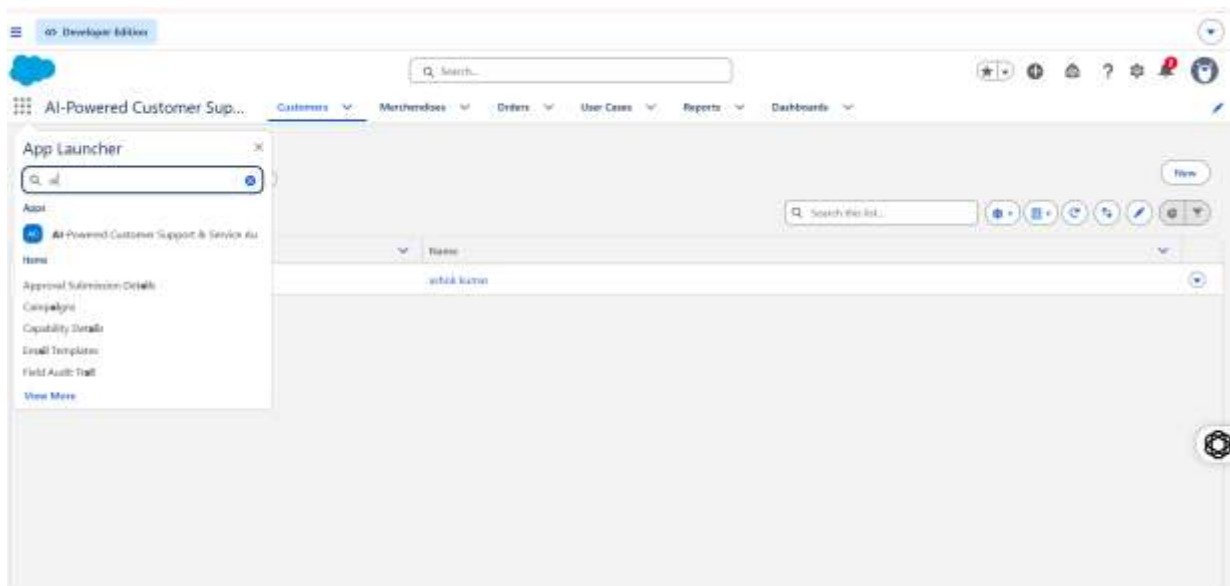
Reports and dashboards are planned to give management ongoing visibility into support performance, including:

- Customers
- Merchendices and Orders

- Case distribution by type (Damage / Warranty)
- Agent performance
- Resolution trends

15.1. Final Application Interface (Project Output)

The final AI-Powered Customer Support & Service Automation application provides a single platform to manage customers, merchandise, orders, and user cases. It integrates Salesforce automation, enabling efficient case management, queue-based assignment, and easy access to reports and dashboards. The application offers a user-friendly interface that improves support operations and reduces manual effort.



16. Testing

- Case creation and record-trigger testing.

- Classification-accuracy testing (Damage / Warranty routing).
- SLA and priority-reassignment testing.
- Validation-rule testing.
- Security and permission testing.

18. Conclusion

The AI-Powered Customer Support & Service Automation project demonstrates how Salesforce Service Cloud, record-triggered Flows, and Agentforce can be combined to modernize customer service operations for an international Electronics & Gadgets company. The solution provides structured case management, automated classification and SLA-driven routing, AI-assisted agent support, secure role-based access, and reporting for continuous improvement — replacing a manual process with a faster, more consistent, and more visible support system.